

ADAPTIVE TIMING SIGNAL CONTROL IN VIETNAM TRAFFIC SYSTEM

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Abstract—Traffic congestion is a significant and challenging problem in Vietnam, resulting in increased waiting times, deteriorating air quality, and higher emissions. However, current fixed-timing traffic light systems are ineffective in managing dynamic traffic flows. In this paper, we propose an adaptive traffic signal control system to address these issues. The system uses a surveillance camera equipped with a YOLO machine learning model to count various vehicle types in real time. By utilizing modified Passenger Car Unit (PCU) values tailored to Vietnamese traffic, this approach dynamically adjusts green light duration to alleviate congestion. The detection model is trained using the up-to-date and diverse UIT-VinaDeveS22 dataset. The expected outcome is to improve traffic flow and reduce congestion at a single intersection, providing a more efficient and responsive solution for Vietnam’s urban traffic challenges. The performance of the system will be evaluated based on some metrics, including Average Vehicle Delay, Queue Length, Throughput, and the standard deviation of PCU scores.

Index Terms—YOLO, Traffic Light, UIT-VinaDeveS22, Timing Signal Control

I. INTRODUCTION

A. Traffic Congestion in the Vietnamese Context

Vietnam is the 15th most populous country in the world. More than 40 million people currently reside in big cities such as Hanoi, Ho Chi Minh City, and Can Tho, and avoiding traffic jams is impossible during peak hours. In Vietnam, the most popular means of transport are the two-wheeled vehicle, followed by the car and other public transport [1]. Although the occupied area of motorbikes is small compared to others, their high number still affects the traffic bottleneck. During peak hours, many drivers do not comply with traffic rules; they suddenly move to another lane that is less crowded or to pedestrian lanes, leading to chaotic situations [2].

Traffic congestion leads to increased energy consumption, significant greenhouse gas emissions, and vehicle noise, resulting in diminished air and sound quality [3], [4]. Moreover, the time loss due to waiting time can cause drivers to lose time and effort, thereby reducing their productivity [5], [6]. It increases the mental issues of citizens due to the long-lasting exposure to vehicle noise, eventually leading to high pressure and depression. Traffic congestion also impacts vehicle equipment, leading to increased fuel consumption and accelerated wear [7], [8]. The crowded street deteriorates quickly than others due to handling long exposure lines of means of transport,

adding maintenance fees on the government and individuals. Eventually, it can cause drivers to incur more transportation costs. Solving the traffic congestion would require developing the traffic management, improving the infrastructure, and the cooperation of the community. One of the main reasons is the traditional traffic light system, which functions based on a fixed timing signal [9], [10], [11].

B. Fixed Timing Signal Control’s Problems

The waiting time of vehicles at traffic lights contributes to the most common traffic congestion problem in Vietnam [12], [13], because it prevents vehicles from moving forward [14]. Currently, most of Vietnam’s traffic lights operate in a traditional fixed-timing mode, which is a tri-bulb traffic light. The two main concepts of traffic signal control are the traffic light cycle and the traffic light phase. A traffic light cycle is a sequence of light signals, with ‘red’ indicating vehicles must stop, ‘green’ indicating vehicle movements, and ‘yellow’ indicating a slowdown. The traffic phase is the duration of each phase of the lane signal. Although the preconfigured traffic signal is suitable and effective for static, sparse traffic, it is not appropriate for dynamic variable flow. This is because the allocated time was often defined based on the traffic engineer’s observation, and the specific traffic studies and data used to control these fixed timings are often not publicly accessible, making it difficult to assess their work in a rapidly changing area [15]. To adapt effectively to traffic situations, traffic signals must depend on real-time traffic flow and volume.

However, while adaptive control methods have been explored, many face challenges in real-world Vietnamese scenarios, often due to their reliance on outdated datasets or complex algorithms that depend on unrealistic assumptions about uniform lane widths [16]. Therefore, a new approach based on Adaptive Timing signal control is an intuitive way to solve this problem to the unique characteristics of Vietnamese traffic. The new approach considers several key factors, including the number of each type of vehicle, modified PCU scores for each type of vehicle, and the lane width value. These values play an important role in calculating the final score for each lane, and based on that calculated value, the extra green time length is given accordingly.

The remainder of the paper is organized as follows: Section III reviews other techniques and approaches; Section IV covers the methodology, including descriptions of the chosen training dataset, the proposed equation to determine the PCU score for each lane, and the traffic workflow to optimize traffic jams; finally, Section V summarizes the paper.

II. RELATED WORK

The Adaptive Timing Signal control method has been researched for many decades and is widely applied in many developed countries. Many studies have been published proposing a similar approach and achieving high performance in simulations. Some methods have focused on optimizing large urban networks with multiple intersections [17]. For instance, in 2021, Wang [18] proposed a model combining Cellular Automata and Genetic Optimization Algorithms, while Wang [19] suggested an Adaptive Multi-Input and Multi-Output Linear Quadratic Regulator (MIMO LQR). Although these methods proved effective in a simulation environment like SUMO, their complexity and reliance on unrealistic assumptions—such as uniform lane width across the entire road body—limit their applicability in real-life scenarios, especially in Vietnam [16]. In addition, the MIMO method requires that the street can be linearized and contains no noise to accommodate to the formula.

In contrast, other research has focused on solutions for single intersections using machine learning and detection techniques. For example, Ali [20] used a Support Vector Machine (SVM) to classify junctions based on congestion level. A more direct approach was proposed by Vishnu [14], which used YOLOv3 cameras to count vehicles and a MATLAB GUI to distribute green time. Similarly, Dung Nguyen [1] adopted a YOLOv8 model combined with a Fuzzy Algorithm to determine green light timing. These approaches, while successful in simulation models, are more suitable for implementation in Vietnam due to their relative simplicity and reasonable cost.

More recent studies have begun to address multimodal and human-centric traffic, a critical aspect of urban traffic. Wang [21] applied a Reinforcement Learning (RL) model to focus on a variety of vehicles and pedestrians, ensuring human-centric equality. Similarly, Yazdani [22] proposed a Deep Reinforcement Learning (DRL) approach to balance vehicle and pedestrian traffic. While these techniques have shown potential for addressing the Vietnamese traffic system, they all use outdated or irrelevant training datasets. These systems depend on general datasets like UTM-Ha Noi, Waymo Open Dataset, KITTI Vision, etc.; therefore, despite their high accuracy in many computer vision applications, they did not accurately reflect the unique vehicle composition, especially the high number of motorcycles, found in Vietnamese traffic. This makes them less effective than a specialized, local dataset such as UIT-VinaDeveS22, and eventually leads to inaccurate counting outputs in real-life scenarios.

Overall, while previous studies have made significant progress in adaptive signal control, a gap remains in combining a sophisticated detection model with an up-to-date



Fig. 1. Some images captured in the dataset UIT-VinaDeveS22

and context-specific dataset for real-world Vietnamese applications. Our proposed approach aims to fill this gap by leveraging the strengths of real-time object detection and a modern, diverse dataset to deliver a more accurate and reliable solution for urban traffic congestion.

III. METHODOLOGY

A. Training dataset UIT-VinaDeveS22

UIT-VinaDeveS22 is considered an up-to-date, diverse dataset that captures Vietnamese traffic. The current dataset was gathered from the Vietnamese Department of Transportation. The dataset contains 1364 images collected using closed-circuit television (CCTV). The resolution of the dataset ranged from 354×650 pixels to 720×1280 pixels [23]. The images were collected under various weather conditions in the streets of Vietnam, whether rainy or clear, and captured various levels of traffic density on the street, from sparse to dense. The dataset also covers a wide range of vehicles, including bicycles, motorcycles, cars, trucks, and vans. There are 15418 objects in total, which are listed in the above table [23]. Motorbikes account for the largest proportion, more than three times the number of cars, and the second largest proportion, proving that motorbikes are still the most popular in Vietnam.

TABLE I
NUMBER OF APPEARANCES OF DIFFERENT VEHICLE TYPES

Class	Number of appearances
Bicycle	233
Motorbike	9458
Car	3967
Bus	343
Van	651
Truck	354
Fire truck	432

UIT-VinaDeveS22 is the current dataset that likely reflects the actual Vietnamese traffic context. By using six cameras to capture different scenes on the street, the dataset can cover a wide view and broad context, supplying truthful and real-life scenarios. Cameras are usually placed on many kinds of roads, such as intersections, three-way or four-way junctions, and crossroads, to collect a comprehensive viewpoint. Although UIT-VinaDeveS22 does not cover extreme weather such as fog, the dataset still provides extensive information because it includes various kinds of weather that are popular in Vietnamese urban cities, and it collects many traffic scenes.

TABLE II
NUMBER OF IMAGES AND OBJECT VEHICLES IN DATASET
UIT-VINADEVES22

Set	Number of images	Number of objects
Training	653	7724
Validation	173	1765
Test	538	5929
Total	1364	15418

B. PCU Parameters

The Passenger Car Unit (PCU) is a universal conversion to convert all different kinds of vehicles into a single standard vehicle, which is a car unit [24]. Conversion is useful for coordinating traffic flow management on heterogeneous roads, which are typical road types in many Vietnamese cities. It also easily manages various types of vehicles. PCU conversion is useful and superior in traffic management to other alternative conversions, and it is widely adopted in many previous adaptive timing methods. Depending on the occupied volume on the road and the speed of the vehicle, the weight will be selected differently to accommodate the current traffic situation. Because this project is based in Vietnamese urban cities, a speed below or equal to 40 km/h is considered. The common PCU is IRC 106:1990, which was established based on the Indian traffic system [24]. To fit into the context of Vietnamese traffic, we selected the Vietnamese traffic vehicle conversion established by the Vietnamese General Department. The table II compares the universal PCU conversion with the Vietnamese conversion.

As observed from the table, overall, the weight of the Vietnamese vehicle conversion is lower than the IRC score in all vehicle types, except for fire trucks. This may be because Vietnamese drivers can easily occupy the space between two vehicles and travel side by side. For the vehicles in different countries, drivers must comply with the traffic law and usually keep the same lane for driving, unless changing directions. Vietnamese traffic drivers can be more flexible and fill in the gaps to move ahead. In summary, the final equation to determine the PCU score for each road is as follows:

$$PCU_{TotalScore} = \sum_{i=1}^n (c_i \cdot w_i) \quad (1)$$

where:

- n : total number of vehicles,
- c_i : number of specific vehicles (motorbikes, cars, buses, bicycles, etc.),
- w_i : corresponding PCU weight value for that type of vehicle.

This equation ensures equality and accuracy in determining the crowdedness of the street, which helps monitor and compute the appropriate bonus green time length in the final task.

C. Analyzed Threshold

The adaptive timing signal control only operates in case one of the lanes whose PCU score is over the analyzed threshold



Fig. 2. Subtasks of Adaptive Timing Signal Control

threshold. The threshold value $t_{threshold}$ indicates that the road is crowded and needs to be released. To determine a suitable threshold to activate the adaptive timing control, it is crucial to investigate the current Vietnamese traffic system, including lane width and allowed speed, the number of each type of vehicle, and ensure that this threshold value can be applied to every Vietnamese road type.

There are several steps that may be considered to determine the threshold value in the simulation world using the SUMO (Simulation of Urban Mobility). First of all, we create the Vietnamese street model in the SUMO system. The model will be filled with vehicles that are listed in the dataset UIT-VinaDeveS22 to ensure the simulation model accurately reflects the Vietnamese situation. Secondly, we run the model several times and use the different threshold values. After each episode, we analyze the quantitative evaluation metrics to estimate the most accurate value that maximizes traffic flow. Finally, based on the obtained result, we can determine the most suitable threshold value.

D. Traffic Workflow

The problem was divided into three subtasks, as illustrated in the fig. 2 above:

In the first and second tasks, the surveillance camera detects vehicles using YOLO technology [8]. YOLO version 8 was selected for this project due to its optimal balance between high detection accuracy and fast processing time [2]. This makes it superior to other YOLO versions in terms of accuracy and lightweight, which might compromise performance. In an experiment of a real-time traffic scenario, YOLO version 8 is proven to be the best due to its high accuracy percentage (up to 88 percent) in the counting and classification tasks [1]. Four cameras were placed on four roads at the intersection to capture the actual number and type of vehicles passing through [25]. The camera was placed at a height of 8 meters and a distance of 200 meters away from the traffic light. The 8 m height was chosen to provide a wide range of observation and minimize vehicle occlusion, a common challenge in Vietnam's high-density traffic. The 200-meter distance ensures that the system can detect and count vehicles well in advance, allowing the adaptive timing to be calculated and implemented before cars come to a complete stop, thus maximizing traffic flow and preventing a "stop-and-go" situation. The camera's counting area was 20 meters long. That area is suitable for analyzing and counting vehicles. It would ensure the counter is accurate and reliable without consuming computational resources. The processing time is not long, so the decision can be made in

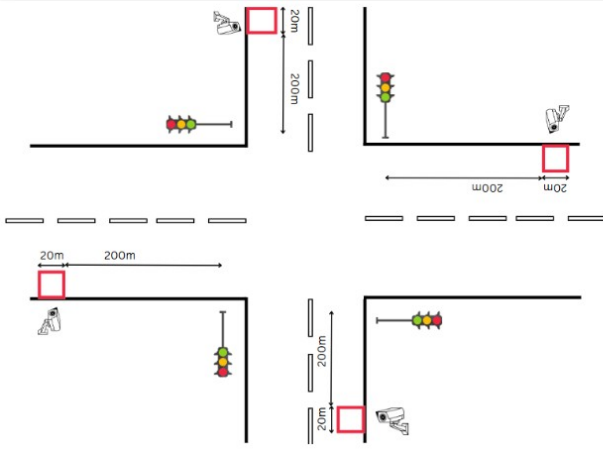


Fig. 3. Graphic Design of Adaptive Timing Signal Control

real-time. Figure 3 shows the comprehensive framework and design of the traffic system. The camera started detecting and counting objects in the last 25 seconds of the green light. This time period is selected based on the 200-meter distance from the traffic light and the speed of vehicles in urban areas. In Vietnam, the maximum allowed speed for all vehicles is 40 km/h, which is about 11 m/s. However, running at maximum speed on a crowded street is not possible, so it is assumed that the average speed is 31 km/h, which is 8.61 m/s. The total distance from the counting area to the traffic light is 220 m. Therefore, the estimated time is:

$$\frac{220 \text{ m}}{8.61 \text{ m/s}} \simeq 25 \text{ (s)} \quad (2)$$

The counting result is transmitted to the controller system via Transmission Control Protocol Modbus (Modbus TCP). Modbus TCP is selected due to its wide application in infrastructure and traffic regulation, as well as its ability to communicate information efficiently. In addition, Modbus protocols function most suitably within a SCADA system [26], [27], [28]. The controller system is a SCADA system equipped with PLCs (Programmable Logic Controllers). SCADA (Supervisory Control and Data Acquisition) is well-known for its effectiveness in controlling, supervising, and collecting information automatically. It has been widely applied in industrial control systems (ICS) for many decades. Its ability makes it the most suitable option in a traffic system. The SCADA system has the PLC as a temporary memory storing the values to be analyzed. The counting variables are updated simultaneously in the PLC memory [29], [30], and SCADA uses these values to calculate the PCU score and assign the bonus green time length.

The extra green light time was calculated based on the difference in the PCU scores of both roads. The equation is as follows:

$$t_{\text{moreGreen}} = \min(k \times |\text{laneWE.score} - \text{laneNS.score}|, T_{\text{max}}) \quad (3)$$

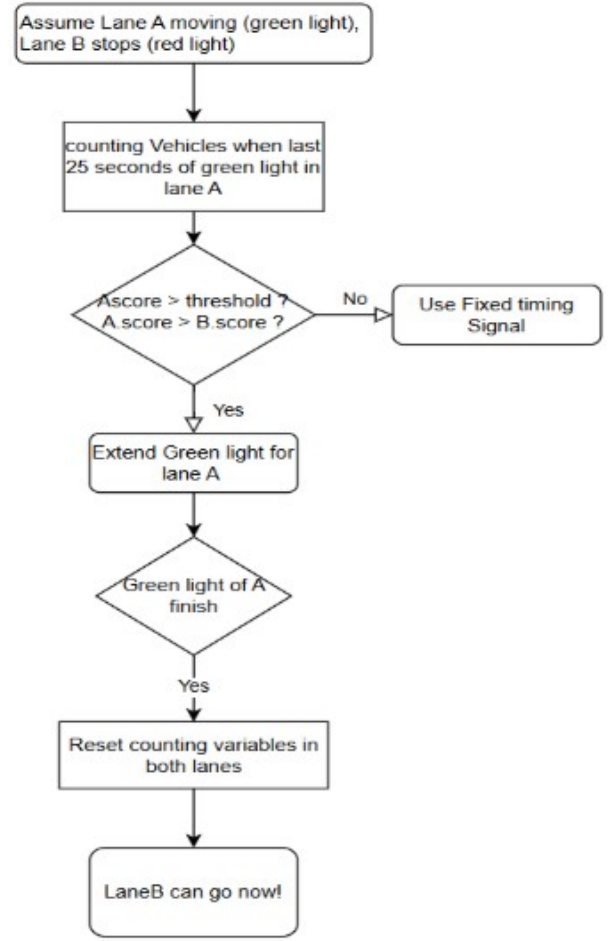


Fig. 4. Workflow of Adaptive Timing Signal Control

where:

- $t_{\text{moreGreen}}$: the bonus green light time
- k : the traffic coefficient
- laneWE.score : PCU score of the West-East Lane
- laneNS.score : PCU score of the North-South Lane
- T_{max} : The maximum extra of green light time

The final task is traffic flow optimization, which is responsible for monitoring and coordinating traffic management. This new method is a combination of the traditional timing mode and an updated approach. For normal times with low traffic density, the fixed timing mode operates adequately. When one of the lanes has a PCU score over the estimated threshold, the adaptive timing is triggered and continues for the rest of the traffic light phase. The new method will solve the issues once the traffic congestion is cleared (PCU score of both lanes is less than the threshold value) and return to the normal operation of the traditional mode. The workflow of this optimization is illustrated in the flowchart fig.4.

This equation ensures equality and accuracy in determining the crowdedness of the street, which helps monitor and compute the appropriate bonus green time length in the final task. Every cycle of traffic lights gives each road at the

intersections one chance to receive extra green time to clear the traffic congestion. It also ensures that the other leg of the intersection does not contain vehicles waiting too long, in case the benefited street always has a long queue length. The coefficient k value and maximum bonus green light time will be determined through online simulation, similar to the process of finding the threshold value.

IV. EVALUATION METRICS

Firstly, we need to evaluate the performance of YOLOv8 to ensure it does not detect and count missing vehicles. The counting variables are increased immediately when the vehicles pass through the boundary line. The cameras will be tested in various weather conditions to ensure their robust ability. Once the expected performance of the counting process is satisfied, we consider the process of classifying objects. The objects need to be classified correctly, so the calculated PCU values are correct for the next phase. In the first two phases, we have to check it manually.

Secondly, the process of regulating traffic flow is evaluated by the standard deviation of time. In the traffic management area, the standard deviation of time is considered as volatility in changes of signal intervals [1], [31]. The lower the variable, the smoother traffic flows and the safer drivers [13], thus we need to minimize this variable. Drivers can easily predict the next timing signal, thereby reducing the waiting time and experiencing a smoother traffic flow. The standard deviation value is obtained by the equation as follows:

$$\sigma = \sqrt{\frac{\sum_{i=1}^N (x_i - \bar{x})^2}{N}} \quad (4)$$

Where:

- N is the total number of times
- x_i is the number of values
- $\bar{x}$ is the average value.

In addition, there are some alternative quantitative metrics that can be used to evaluate the timing signal. Firstly, the average vehicle waiting time will be used to measure how long vehicles are delayed at the intersection. The time duration is from when the vehicle passes the starting boundary line to when it exits the intersection. A smaller value indicates that traffic flow is optimized and the driver waits less time. The second metric is the maximum queue length. This value is to measure the buildup of traffic, which evaluates if the approach can solve the bottleneck problem. A lower value of queue length indicates the traffic is smoothly optimized. Finally, we can consider the number of vehicles passing through the intersection during time periods. The high value proves the system works effectively. These quantifier metrics are the basic objects to evaluate the system's performance. However, all of these new metrics are required to install additional cameras at the traffic lights in each junction to detect and count the number of vehicles.

V. CONCLUSION

This study aims to solve the challenging problem of traffic congestion in urban cities in Vietnam by proposing an adaptive timing signal control system. Current fixed-timing traffic light systems are ineffective in managing dynamic traffic flows, leading to increased waiting times, energy consumption, and environmental pollution.

The proposed system utilizes surveillance cameras equipped with the YOLO version 8 machine-learning model to detect and count various vehicle types in real time. The adaptive timing system is designed to operate when a lane's PCU score exceeds a predefined threshold, dynamically adjusting the green light duration to alleviate congestion. By combining real-time vehicle detection, Vietnam-specific PCU parameters, and a responsive control mechanism, this project proposes a solution for mitigating traffic congestion, improving traffic flow efficiency, and ultimately enhancing the quality of life in Vietnamese cities.

While this proposal outlines a robust methodology, the next phase of this project will focus on the real-world implementation and empirical validation of the system. Future work will include conducting a pilot study at a selected intersection in a major Vietnamese city to calibrate the traffic coefficient (k) and determine an optimal threshold value ($t_{\text{threshold}}$) based on real-time traffic data. We also plan to evaluate the system's performance using metrics such as standard deviation value, average vehicle waiting time, queue length, and vehicle throughput. It is worth considering that fixing one intersection might just move the traffic jam somewhere else nearby. Therefore, in the future, we should investigate how this system could work across many intersections, coordinating the lights to keep traffic flowing smoothly everywhere, not just in one spot. We could even explore how this single-intersection system could connect with a bigger network, using a communication system to manage traffic flow over a larger area.

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